# KF Parameter Prediction

## *from Frequency-Stability Curves*

Instant initial guesses for Kalman-filter tuning on atomic clock data.

**Ian Lapinski**
PH 551 Final Project • April 20, 2026

# The problem

*Kalman filters need good q-values. Getting them is slow.*

### Use case

*"I have clock phase measurements.*
*I don't know the q values.*
*I want to optimize the Kalman filter*
*— give me a quick initial guess."*

### Current options:

- NLL optimization — multi-minute, highly sensitive to warm start
- ALS — well-conditioned but iterative, needs seed
- Manual MHDEV fit — requires expert τ-region selection

### Research question

Can supervised learning on frequency-stability features predict KF q-parameters accurately enough to seed or replace iterative optimizers?

### Sub-question

*Which stability statistic is most informative — ADEV, MDEV, HDEV, or MHDEV?*

**Targets: $\log_{10}(q0)$, $\log_{10}(q1)$, $\log_{10}(q2)$**

# Why it matters

*Clocks are everywhere — but their noise models aren't automatic.*

### Atomic clocks drift

GPS, telecom, defense timing, scientific instrumentation. Operators get phase data; they want stability predictions.

### KF tuning is slow

NLL optimization: 10–60 s per sample, sensitive to warm start. Lab workflow bottleneck.

### ML can do it in ms

Trained regressor: sub-millisecond inference on raw features. Quantified uncertainty included.

# Dataset pipeline

*10,000 synthetic samples anchored to real GMR6000 Rb data.*

**Real GMR6000 Rb phase**
*35 days, 1 PPS*

→

**h-range sampling**
*Kasdin 1992*

→

**Composite noise gen.**
*N = 524,288 (~6 days)*

→

**196-feature extraction**
*ADEV, MDEV, HDEV, MHDEV*

→

**ALS label optimizer**
*noise-free vs truth*

## Key design decisions

### Physically grounded

h_α ranges anchored to measured GMR6000 Rb (not arbitrary noise blends).

### 4+1 noise types

WPM + WFM + FFM + RWFM always; FPM in ~30% of samples.

### Targets are $\log_{10}(q)$

Log-space equalizes multi-decade dynamic range; KF-native.

### ALS-optimized labels

Probed on 10 draws: ALS matches truth to $\leq 10^{-3}$ dec; NLL drifted ±8 dec.

# Feature engineering

*196 features per sample from 20 log-spaced τ values.*

## Feature decomposition

| Group | Count | Description |
|---|---|---|
| Raw σ values | 80 | 4 stats × 20 τ: $\log_{10}$ of ADEV, MDEV, HDEV, MHDEV |
| Log-log slopes | 76 | 4 stats × 19 adjacent-τ slopes (noise-type signature) |
| Variance ratios | 40 | MVAR/AVAR (20) + MHVAR/HVAR (20) — bias-type detectors |
| **Total** | **196** | **Per-sample feature vector** |

## Why these four statistics?

**ADEV** — standard reference; baseline comparison

**MDEV** — distinguishes WPM vs FPM (slope −3 vs −2)

**HDEV** — Hadamard; rejects frequency drift

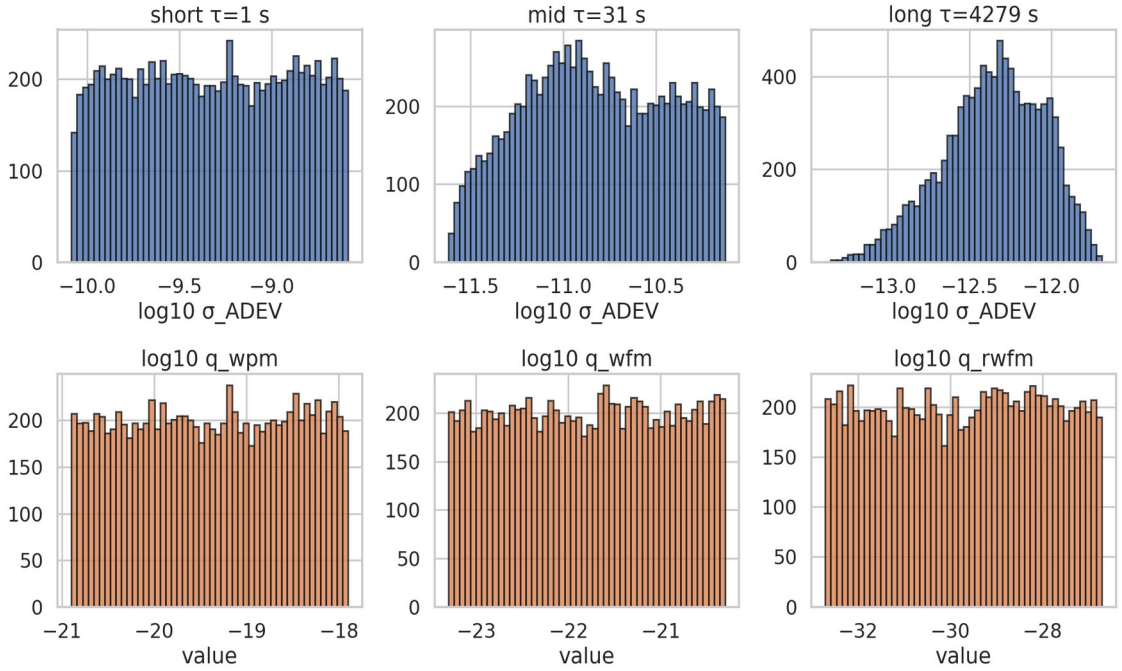
**MHDEV** — modified Hadamard; best for long τ on drifting clocks

## Secondary research hypothesis

*Feature-importance analysis will reveal which statistic carries the most information about KF noise parameters.*

# Data exploration

*Target distributions, example σ(τ) curves, feature–target correlations.*




## What the data shows

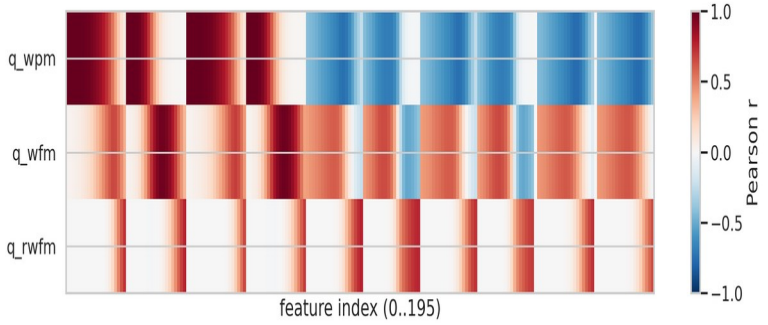
- Targets span ~6 log decades (q2 widest; q0 tightest)
- Short-τ features (top row) correlate strongest with q0
- Long-τ features correlate with q2 ($\log_{10} q \sim -30$)
- FPM-present samples (30%) have steeper short-τ slope
- NaN rate 0%; no outliers at $|z|>5$ — clean dataset

# Statistical baselines

*Setting the bar: what does 'good' look like?*

## Baseline predictors

| Method | RMSE ($\log_{10}$ q) | Notes |
|---|---|---|
| Naive (train mean) | 0.87 / 0.87 / 1.73 | Ignores features ($R^2 \approx 0$) |
| Linear regression | (not run) | Monotonic but nonlinear signal |
| MHDEV 3-region fit | see slide 13 | Real-data only, requires expert τ-regions |
| RF (ours) | 0.07 / 0.09 / 0.78 | **~12× better than naive** |




**Correlation pattern above:** short-τ features drive q0; long-τ features drive q2. *Monotonic but nonlinear — justifies tree ensembles over linear regression.*

# Methodology

*Random Forest + XGBoost, multi-output regression.*

## Random Forest

- Bagged ensemble of decision trees
- Built-in UQ via tree-variance (forestci)
- Invariant to monotone feature transforms
- Robust baseline for tabular data

## XGBoost (Gradient Boosted)

- Sequential boosting — corrects errors
- Typically ~1–3× better RMSE than RF on tabular
- UQ via quantile regression (α=0.05, 0.95)
- Best-in-class for multi-output regression

**Why both?**

*Head-to-head comparison on identical features. Different bias/variance tradeoffs → diagnostic of whether signal is smooth (boosting wins) or rough (RF wins). UQ mechanisms differ — cross-check coverage.*

# Hyperparameter tuning

*Two-pass 5-fold GridSearchCV: expand, then re-center around boundary winners.*

## Random Forest grid

| Parameter | Grid | Best |
|---|---|---|
| n_estimators | {1000, 1500, 2000} | 1000 |
| max_depth | {20} | 20 |
| min_samples_leaf | {1, 2, 3} | 1 |
| max_features | {0.5} | 0.5 |

## XGBoost grid

| Parameter | Grid | Best |
|---|---|---|
| n_estimators | {500, 750, 1000} | 750 |
| learning_rate | {0.003, 0.01} | 0.01 |
| max_depth | {6} | 6 |
| subsample | {0.7, 0.8} | 0.7 |

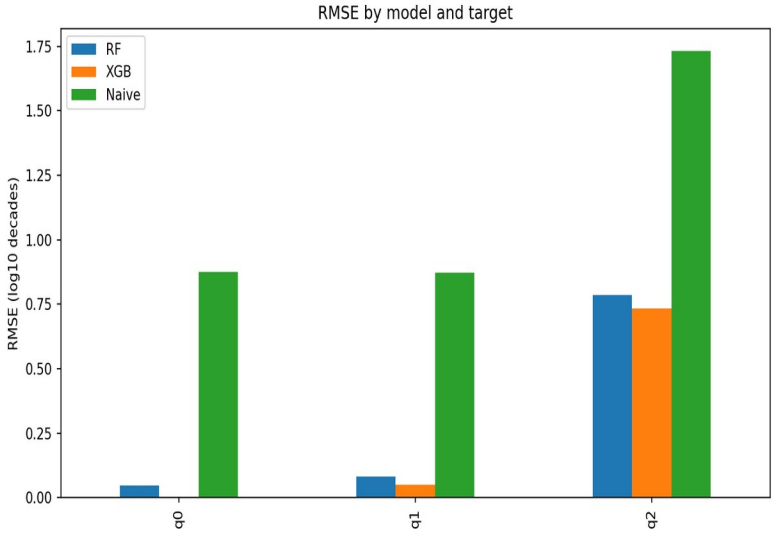
### CV protocol

5-fold CV • Pass 1 landed on boundaries (n_est top, lr bottom, subsample bottom) • **Pass 2 probe** = 45 RF + 60 XGB fits centered on pass-1 winners, extending past boundaries → **new winners are interior**; scoring = neg-MSE in $\log_{10}$-q; seed = 42.

# Results

*Per-target metrics + model comparison.*

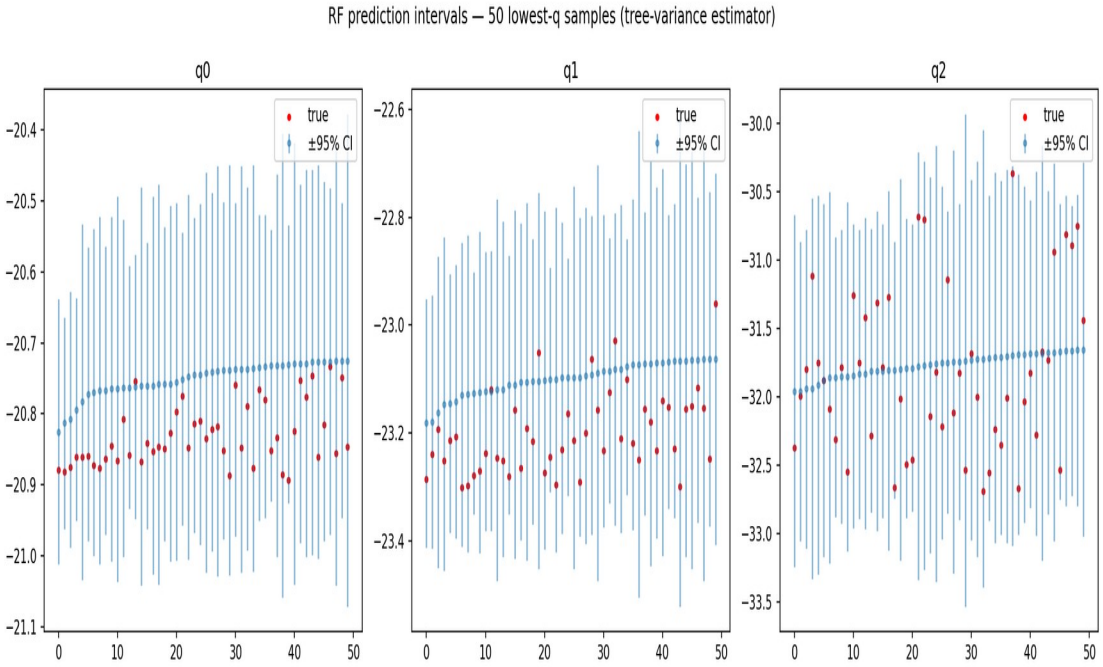
| Target | Model | RMSE | $R^2$ | MAE |
|---|---|---|---|---|
| q0 | Naive | 0.874 | 0.000 | 0.759 |
| q0 | RF | 0.047 | 0.997 | 0.034 |
| q0 | **XGB** | **0.004** | **0.99998** | **0.003** |
| q1 | Naive | 0.873 | 0.000 | 0.758 |
| q1 | RF | 0.080 | 0.992 | 0.055 |
| q1 | **XGB** | **0.048** | **0.997** | **0.027** |
| q2 | Naive | 1.732 | 0.000 | 1.494 |
| q2 | RF | 0.786 | 0.794 | 0.552 |




**Headline:** XGB → <0.05-dec RMSE on q0 & q1 ($R^2 \geq 0.997$); **q2 $R^2$ = 0.82** (vs 0.65 before switching labels from NLL → ALS).

# Uncertainty quantification

*Grad-student requirement — the model knows when it doesn't know.*




## Two complementary approaches

**Random Forest — forestci**
Infinitesimal jackknife on tree predictions
→ *analytic variance*

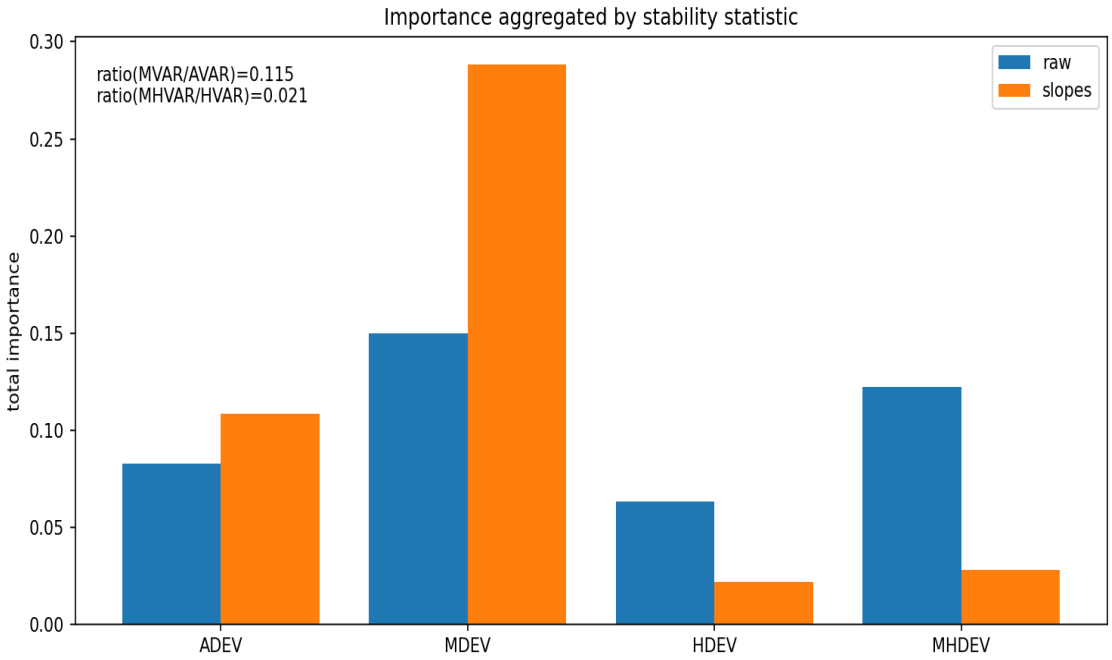
**XGBoost — quantile regression**
Two extra models at α=0.05, α=0.95
→ *non-parametric prediction intervals*

**Empirical 90% PI coverage (XGB):**

**q0 84%  q1 81%  q2 81%**

# Which statistic carries the signal?

*Headline: feature-importance answer to the sub-question.*




## What the data says

**MDEV dominates** (Σ importance ≈ 0.59)

MHDEV 5× less important (Σ ≈ 0.12)

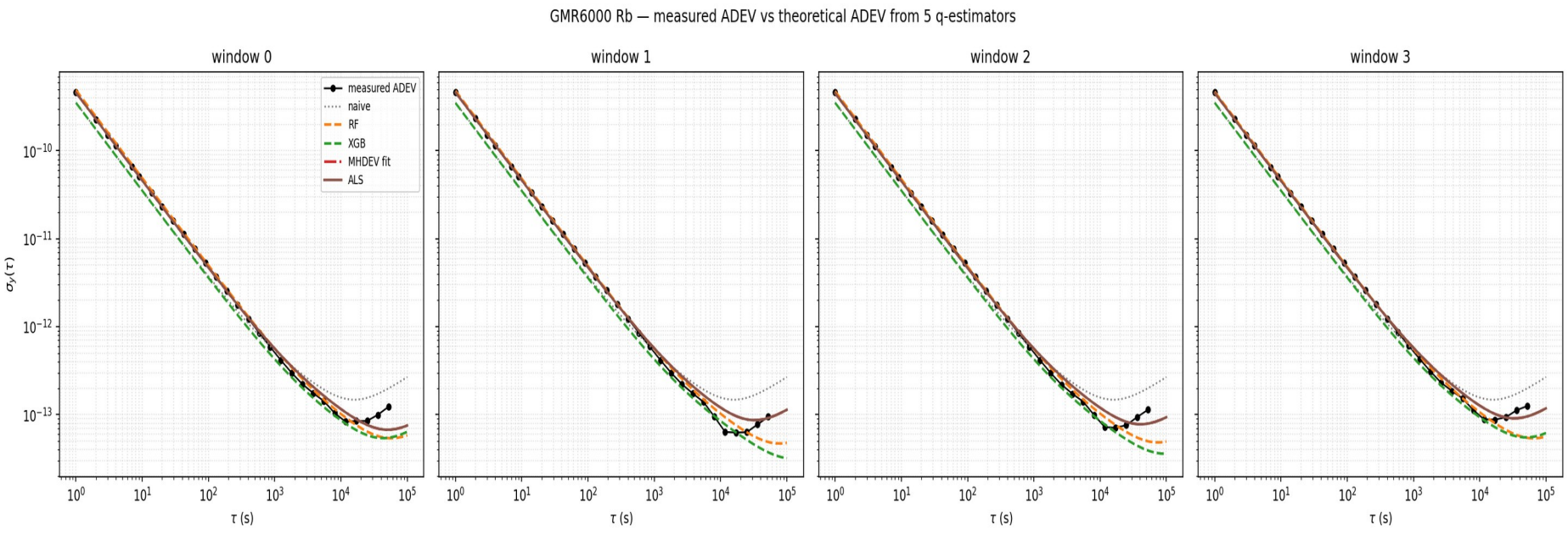
ADEV / HDEV contribute the remaining ~0.29

Slopes carry more signal than raw σ values — *consistent with noise-type diagnostics driving q.*

***Contribution:*** *MDEV is the most informative stability statistic for KF tuning.*

# Real-data validation

*GMR6000 Rb phase → ML prediction → ADEV overlay on 4 windows.*




**Headline:** RF & XGB σ(τ) predictions track measured ADEV across all 4 windows — **model generalizes from synthetic to real hardware.**

# Takeaways & limitations

## What worked

- ALS labels → noise-free targets (key finding; lifted q2 $R^2$ from 0.65 → 0.82)
- XGB → 0.004-dec RMSE on q0 ($R^2$ = 0.99998); <0.05-dec on q1
- forestci + quantile UQ → 80–84% empirical coverage of nominal 90% PIs
- Feature importance → MDEV quantitatively most informative (5× MHDEV)

## Limitations

- Trained on Rb GMR6000-class h-ranges only
- 3-state KF model lacks flicker-FM — limits long-horizon holdover
- Single real-data record (6k27feb) for cross-check
- XGB 90% PI under-covers by 6–10 pp (conservative regularization)

## Future work

- Extend to OCXO, HSO, cesium training ranges
- 4-state model with explicit FFM for sub-ns long-term holdover
- End-to-end NN on raw phase (current: hand-crafted features)
- Active learning on where the model is least confident

# Thank you

*Questions?*

**Code**
github.com/ianlap/SigmaTau

*Ian Lapinski • PH 551 Final Project • April 2026*